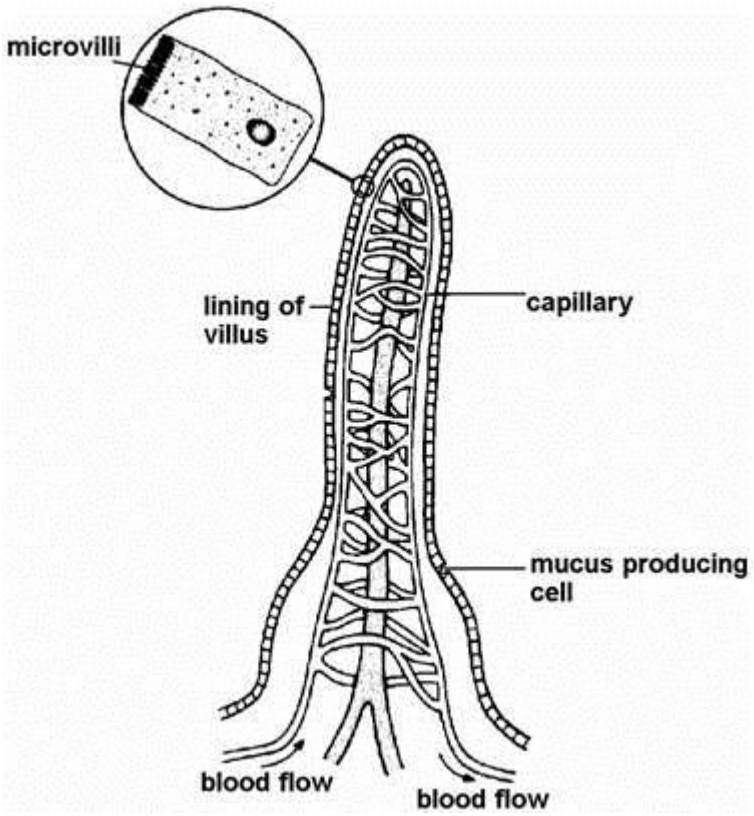
Q1.

The diagram below shows a villus. There are several thousand villi in every square centimetre of the lining of the small intestine. The cells in the lining of each villus have folds called microvilli on their outer surfaces. One of these cells is shown magnified in the diagram.



(a) Explain **two** ways in which the structure of the villus enables the products of digestion to be absorbed efficiently.

1. .....

.....

2. .....

.....

2 marks

(b) Mucus is produced by some cells in the lining of the small intestine.
One function of mucus is to protect the lining of the small intestine against the action of certain chemicals.

(i) Give the name of **one** type of chemical present in the small intestine which would otherwise damage the lining.

.....

1 mark

(ii) Suggest **one other** function of mucus produced in the small intestine.

.....

.....

1 mark

(c) Bile, produced by the liver, enters the small intestine through the bile duct. Describe the effect which bile has on fats in the small intestine.

.....

.....

1 mark

Maximum 5 marks

Q1.

(a) any **two** from

- it **or** the villus has a large surface
  accept 'they have a large surface area'
- the microvilli have a large surface area
- it has a good blood supply **or** lots of capillaries
- the lining is one cell thick
  *accept 'the lining is thin' **or** 'the blood is close to the surface' **or** 'the blood vessels are close to the surface'*

2

(b) (i) enzymes
*accept 'acid' **or** 'alkali' **or** 'HCl' **or** 'protease'*

1

(ii) any **one** from

- lubricates the food
- makes food pass along the intestine easily
- prevents food sticking in the intestine
  accept 'it is part of the immune system'

1

(c) any **one** from

- emulsifies them
  accept 'emulsifies it'
- breaks down large globules into smaller globules
  *do **not** accept 'digests fat'*
  ***or** 'breaks it into smaller molecules'*
  ***or** 'breaks them down'*
- increases the surface area

1

[5]